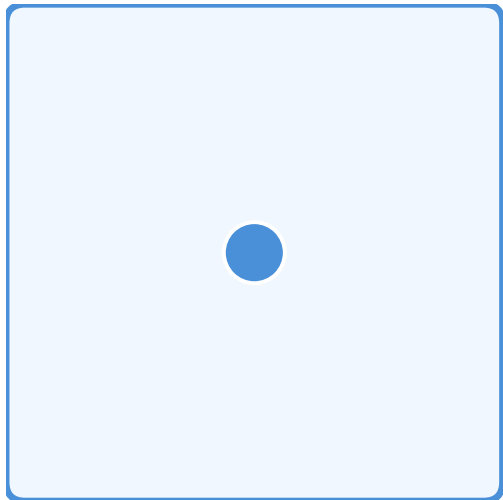


Training data selection by reasoning topology (Besta et al., 2024)

Atomic

$|V| = 1$



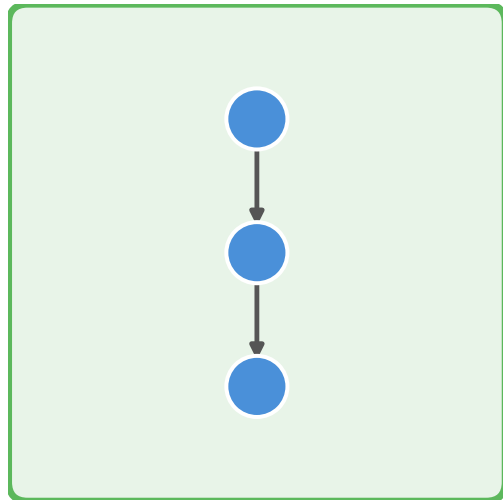
When not to
decompose

GSM8K

n = 500

Path

Linear chain, $\text{deg} \leq 1$



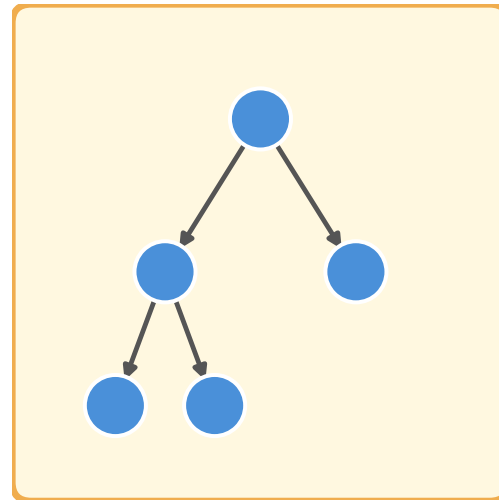
Sequential
delegation

NuminaMath, HotpotQA

n = 1.5k each

Tree

Allow branching ($\text{out-deg} > 1$)



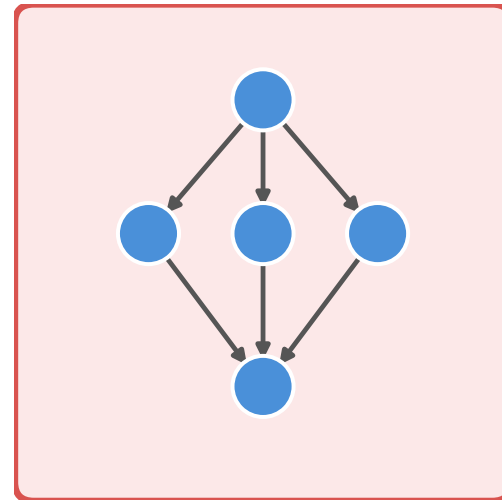
Explore alternative
decomposition paths

MuSiQue

n = 1.5k

DAG

Allow fan-in ($\text{in-deg} > 1$)



Parallel sub-tasks +
tool chaining

DROP, TACO, ToolACE

n = 1.5k + 1.75k each

Increasing topological complexity →